ULX5M-M2 v001
Intergalaktik d.o.o.
License: CERN-OHL-S v2
Year: 2026

config
nLED_act
{GPIO}
File: config.kicad_sch

power
File: power.kicad_sch

gpdi-ram
File: ram.kicad_sch

Add RP2040 so that we do not need externa JTAG

Add inline resistors for input signals

ethernet
USB_CC1
USB_CC2
{SD}
{USB}
File: ethernet.kicad_sch

4HIGH bit SD

File: lvds.kicad_sch
lvds
{LVDS0}
{LVDS1}
PERST#
PEWAKE#
CLKREQ#
PEDET
SUSCLK
DEVSLP
LED1#

M.2 M Key
PERST#
PEWAKE#
CLKREQ#
PEDET
SUSCLK
DEVSLP
LED1#
{PCIe}
File: ../GateMate_M.2_M_Key/GateMate_M.2_M_Key/M.2 M Key.kicad_sch

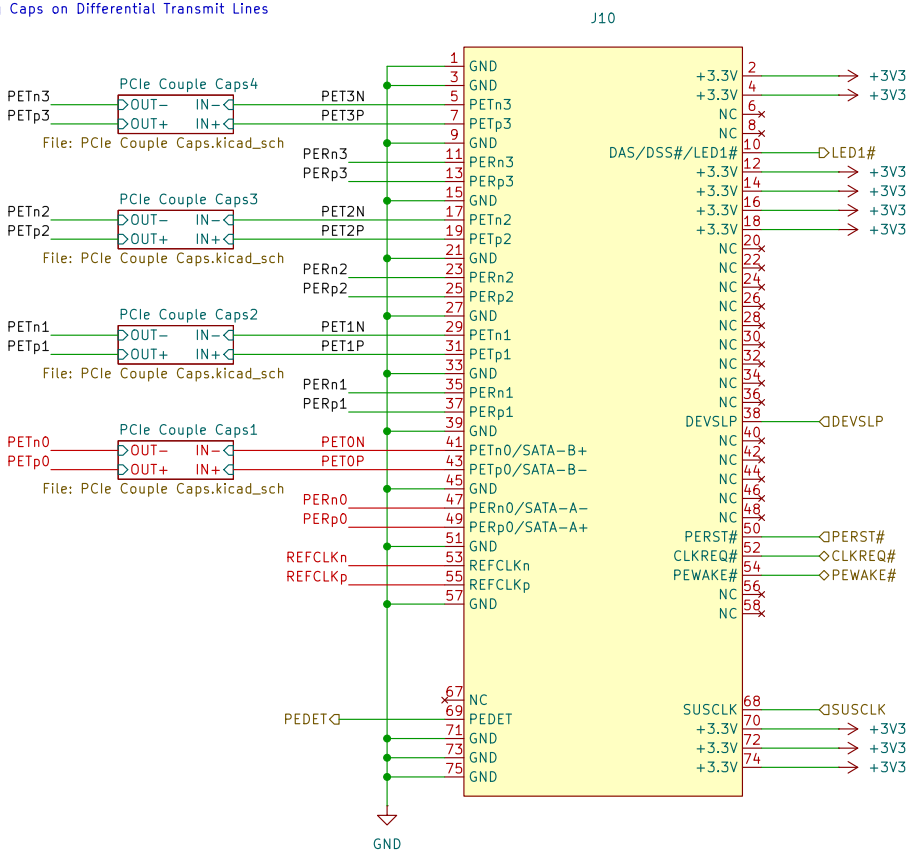
DF40C-100DP-0.4V_51_

serdes
{PCIe}
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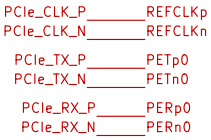
FID104 Fiducial
FID105 Fiducial
FID106 Fiducial
FID101 Fiducial
FID102 Fiducial
FID103 Fiducial

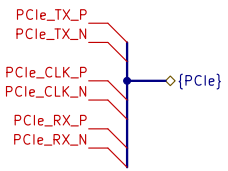
CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /
File: ulx5m-m2.kicad_sch
Title: ULX5M-M2
Size: A4 | Date: 2025-06-24 | **Rev: v001**
KiCad E.D.A. 10.0.6 | Id: 1/12

Put AC Coupling Caps on Differential Transmit Lines





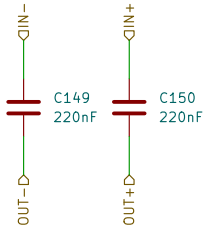




CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /M.2 M Key/
File: M.2 M Key.kicad_sch

Title: ULX5M-M2

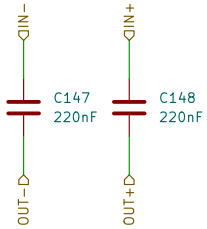
Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 2/12



CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /M.2 M Key/PCIe Couple Caps4/
File: PCIe Couple Caps.kicad_sch

Title: ULX5M-M2

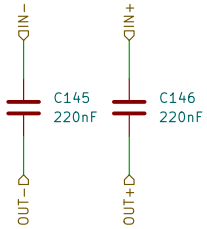
Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 3/12



CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /M.2 M Key/PCIe Couple Caps3/
File: PCIe Couple Caps.kicad_sch

Title: ULX5M-M2

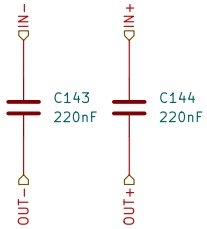
Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 11/12



CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /M.2 M Key/PCIe Couple Caps2/
File: PCIe Couple Caps.kicad_sch

Title: ULX5M-M2

Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 12/12



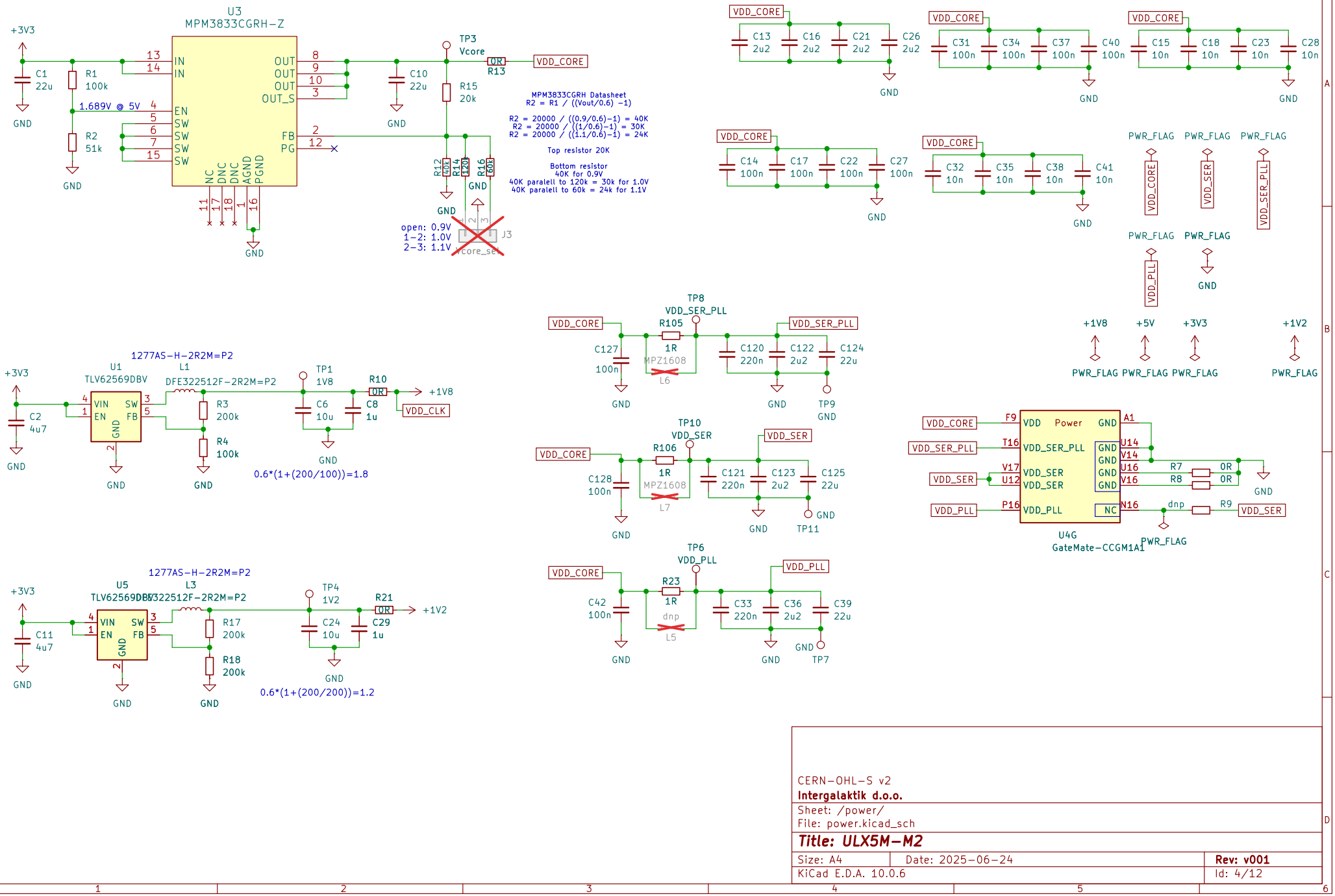
CERN-OHL-S v2

Intergalaktik d.o.o.

Sheet: /M.2 M Key/PCIe Couple Caps1/
File: PCIe Couple Caps.kicad_sch

Title: ULX5M-M2

Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 13/12


U3
MPM3833CGRH-Z
+3V3
C1 22u
R1 100k
1.689V @ 5V
R2 51k
GND
IN
EN
SW
OUT
OUT_S
FB
PG
NC
DNC
AGND
PGND
C10 22u
TP3 Vcore
0R R13
VDD_CORE
R15 20k
R12 40k
R14 120
R16 60k
J3 Vcore_sel
open: 0.9V
1-2: 1.0V
2-3: 1.1V
MPM3833CGRH Datasheet
R2 = R1 / ((Vout/0.6) -1)
R2 = 20000 / ((0.9/0.6)-1) = 40K
R2 = 20000 / ((1/0.6)-1) = 30K
R2 = 20000 / ((1.1/0.6)-1) = 24K
Top resistor 20K
Bottom resistor
40K for 0.9V
40K paralell to 120k = 30k for 1.0V
40K paralell to 60k = 24k for 1.1V
VDD_CORE
C13 2u2
C16 2u2
C21 2u2
C26 2u2
C31 100n
C34 100n
C37 100n
C40 100n
C15 10n
C18 10n
C23 10n
C28 10n
C14 100n
C17 100n
C22 100n
C27 100n
C32 10n
C35 10n
C38 10n
C41 10n
PWR_FLAG
VDD_SER
VDD_SER_PLL
VDD_PLL
+1V8
+5V
+3V3
+1V2
1277AS-H-2R2M=P2
U1
TLV62569DBV
L1
DFE322512F-2R2M=P2
C2 4u7
R3 200k
R4 100k
TP1 1V8
R10
C6 10u
C8 1u
VDD_CLK
0.6*(1+(200/100))=1.8
U5
TLV62569DBV
L3
DFE322512F-2R2M=P2
C11 4u7
R17 200k
R18 200k
TP4 1V2
R21
C24 10u
C29 1u
0.6*(1+(200/200))=1.2
TP8
VDD_SER_PLL
R105 1R
MPZ1608
L6
C127 100n
C120 220n
C122 2u2
C124 22u
TP9
TP10
VDD_SER
R106 1R
MPZ1608
L7
C128 100n
C121 220n
C123 2u2
C125 22u
TP11
TP6
VDD_PLL
R23 1R
dnp
L5
C42 100n
C33 220n
C36 2u2
C39 22u
TP7
U4G
GateMate-CCGM1A1
Power
VDD
VDD_SER_PLL
VDD_SER
VDD_PLL
GND
NC
F9
T16
V17
U12
P16
A1
U14
V14
U16
V16
N16
R7 0R
R8 0R
R9 dnp
CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /power/
File: power.kicad_sch
Title: ULX5M-M2
Size: A4
Date: 2025-06-24
Rev: v001
KiCad E.D.A. 10.0.6
Id: 4/12



Use MD0/1 PU/PD for CPHA/CPOL selection

SW1 219-2MST

U4B GateMate-CCGM1A1

U6 74LVC1G04GM,115

U7 74LVC2G00HK3-7

U8 74LVC2G00HK3-7

U9 AT25QL321-MBUE-T

J101 JTAG_EXT

J102 3V3_PSU

J4 JTAG

CERN-OHL-S v2

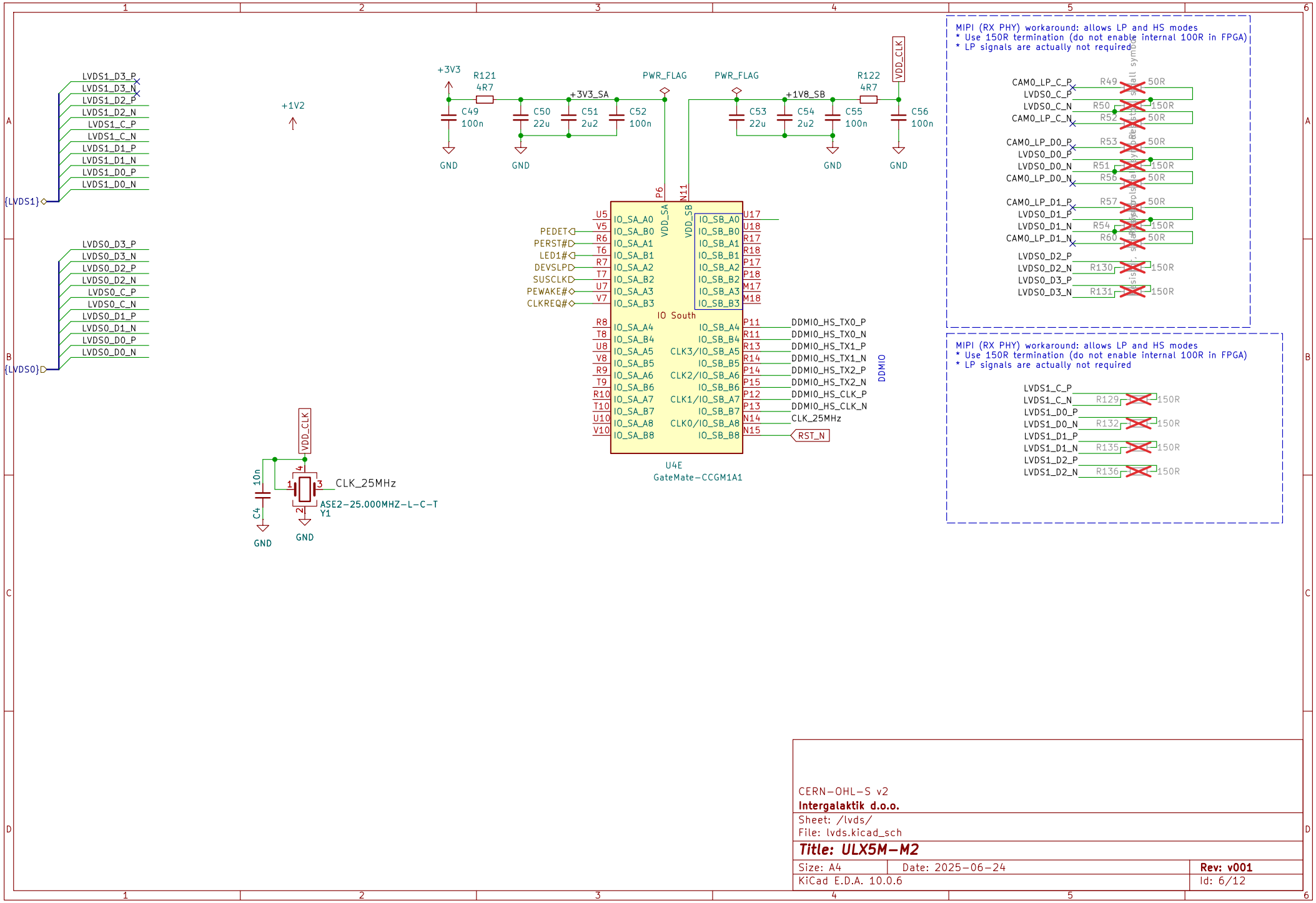
Intergalaktik d.o.o.

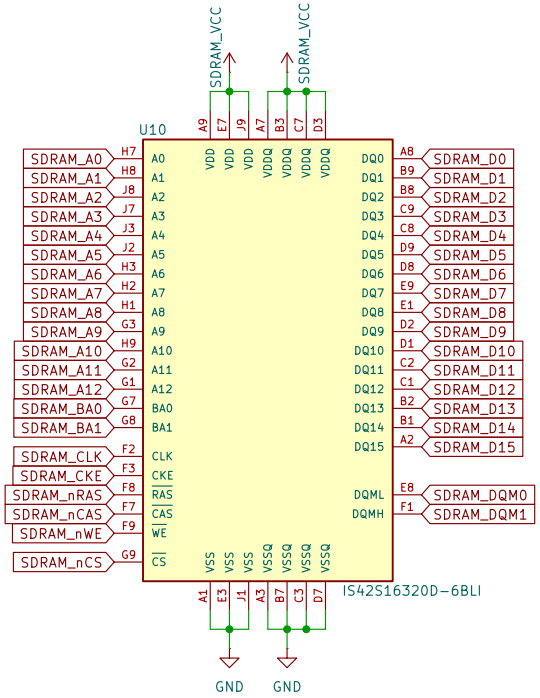
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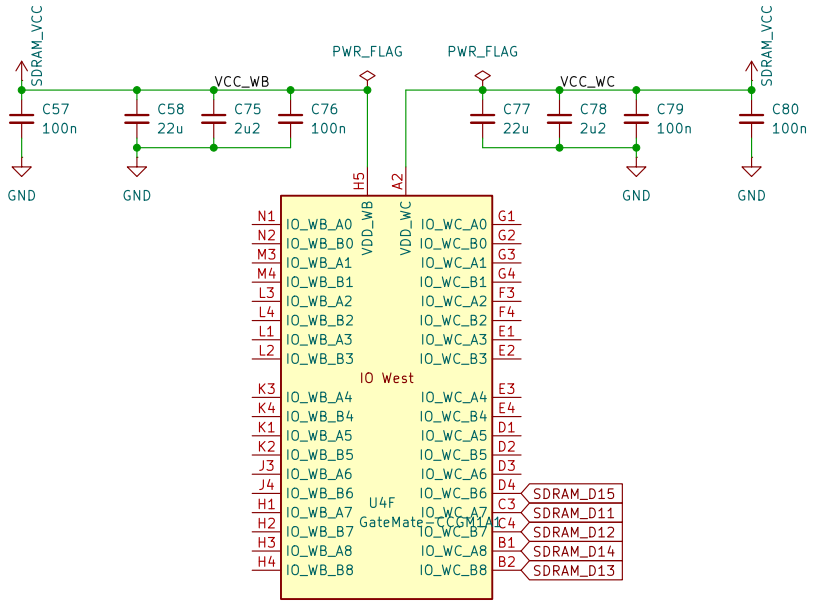
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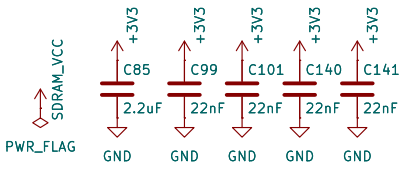
Title: ULX5M-M2

Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 5/12


LVDS1_D3_P
LVDS1_D3_N
LVDS1_D2_P
LVDS1_D2_N
LVDS1_C_P
LVDS1_C_N
LVDS1_D1_P
LVDS1_D1_N
LVDS1_D0_P
LVDS1_D0_N
{LVDS1}
LVDS0_D3_P
LVDS0_D3_N
LVDS0_D2_P
LVDS0_D2_N
LVDS0_C_P
LVDS0_C_N
LVDS0_D1_P
LVDS0_D1_N
LVDS0_D0_P
LVDS0_D0_N
{LVDS0}
+1V2
+3V3
R121
4R7
C49
100n
C50
22u
+3V3_SA
C51
2u2
C52
100n
GND
PWR_FLAG
PWR_FLAG
+1V8_SB
R122
4R7
VDD_CLK
C53
22u
C54
2u2
C55
100n
C56
100n
P6
N11
VDD_SA
VDD_SB
U5 IO_SA_A0
V5 IO_SA_B0
R6 IO_SA_A1
T6 IO_SA_B1
R7 IO_SA_A2
T7 IO_SA_B2
U7 IO_SA_A3
V7 IO_SA_B3
PEDET
PERST#
LED1#
DEVSLP
SUSCLK
PEWAKE#
CLKREQ#
IO_SB_A0 U17
IO_SB_B0 U18
IO_SB_A1 R17
IO_SB_B1 R18
IO_SB_A2 P17
IO_SB_B2 P18
IO_SB_A3 M17
IO_SB_B3 M18
IO South
R8 IO_SA_A4
T8 IO_SA_B4
U8 IO_SA_A5
V8 IO_SA_B5
R9 IO_SA_A6
T9 IO_SA_B6
R10 IO_SA_A7
T10 IO_SA_B7
U10 IO_SA_A8
V10 IO_SA_B8
IO_SB_A4 P11 DDMIO_HS_TX0_P
IO_SB_B4 R11 DDMIO_HS_TX0_N
CLK3/IO_SB_A5 R13 DDMIO_HS_TX1_P
IO_SB_B5 R14 DDMIO_HS_TX1_N
CLK2/IO_SB_A6 P14 DDMIO_HS_TX2_P
IO_SB_B6 P15 DDMIO_HS_TX2_N
CLK1/IO_SB_A7 P12 DDMIO_HS_CLK_P
IO_SB_B7 P13 DDMIO_HS_CLK_N
IO_SB_A8 N14 CLK_25MHz
IO_SB_B8 N15 RST_N
DDMIO
U4E
GateMate-CCGM1A1
VDD_CLK
C4
10n
GND
CLK_25MHz
ASE2-25.000MHZ-L-C-T
Y1
GND
MIPI (RX PHY) workaround: allows LP and HS modes
* Use 150R termination (do not enable internal 100R in FPGA)
* LP signals are actually not required
CAM0_LP_C_P R49 50R
LVDS0_C_P
LVDS0_C_N R50 150R
CAM0_LP_C_N R52 50R
CAM0_LP_D0_P R53 50R
LVDS0_D0_P
LVDS0_D0_N R51 150R
CAM0_LP_D0_N R56 50R
CAM0_LP_D1_P R57 50R
LVDS0_D1_P
LVDS0_D1_N R54 150R
CAM0_LP_D1_N R60 50R
LVDS0_D2_P
LVDS0_D2_N R130 150R
LVDS0_D3_P
LVDS0_D3_N R131 150R
MIPI (RX PHY) workaround: allows LP and HS modes
* Use 150R termination (do not enable internal 100R in FPGA)
* LP signals are actually not required
LVDS1_C_P
LVDS1_C_N R129 150R
LVDS1_D0_P
LVDS1_D0_N R132 150R
LVDS1_D1_P
LVDS1_D1_N R135 150R
LVDS1_D2_P
LVDS1_D2_N R136 150R
CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /lvds/
File: lvds.kicad_sch
Title: ULX5M-M2
Size: A4
Date: 2025-06-24
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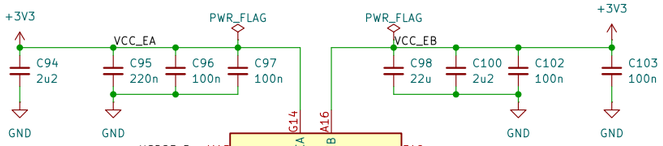


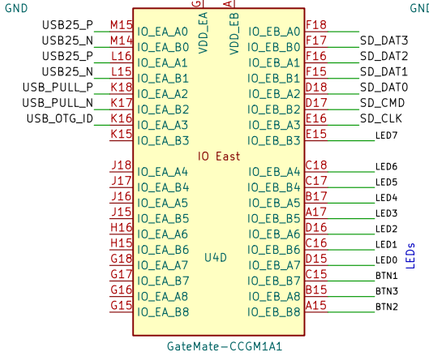
CERN-OHL-S v2
Intergalaktik d.o.o.

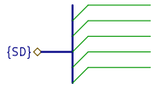
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File: ram.kicad_sch

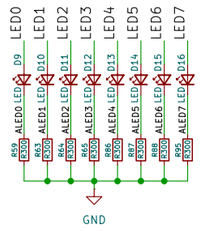
Title: ULX5M-M2

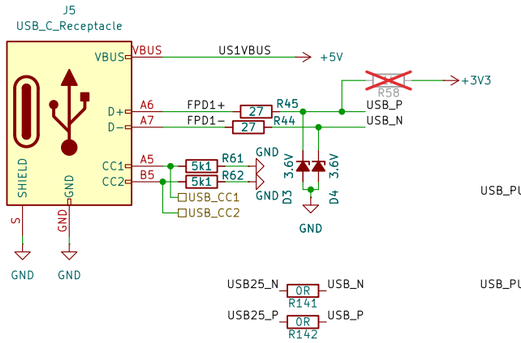
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KiCad E.D.A. 10.0.6		Id: 7/12

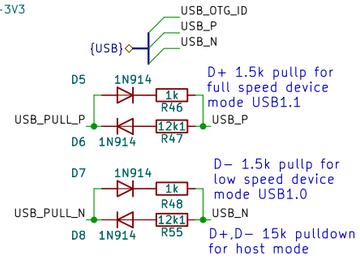


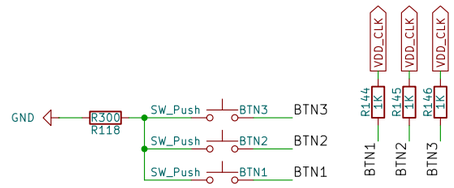








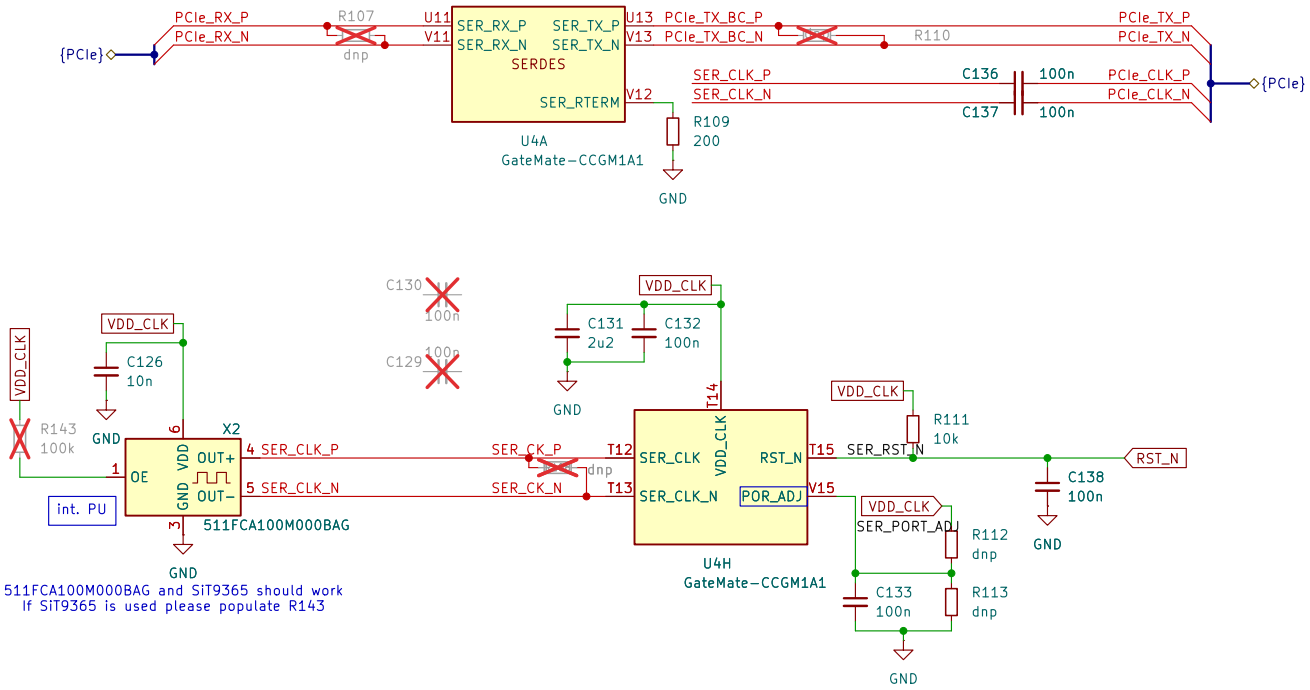




CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /ethernet/
File: ethernet.kicad_sch

Title: ULX5M-M2

Size: A3	Date: 2025-06-24	**Rev: v001**
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Intergalaktik d.o.o.

Sheet: /serdes/

File: serdes.kicad_sch

Title: ULX5M-M2

Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 10/12